The question

#245 produced per-protein contact scores for nine predictors over 314 natural FoldBench monomers, and never explained them. On eval-test every predictor's scores run from 0 to 1. What property of a protein decides where it lands — and is it the same property for MarinFold as for the baselines?

The comparative half is the useful half. If our accuracy tracks training-set proximity while ESMFold2's tracks MSA depth, that says something concrete about what a from-scratch sequence LM has learned, and it tells us which proteins to add to training.

60 features over four families: size and shape, contact geometry, secondary structure, homology and family abundance, plus domains, function, localisation and taxonomy from RCSB and UniProt. Analysis only — every score already existed.

One answer, for every predictor

Family abundance. How many relatives a protein has — MSA depth, homologs in our corpus, KNN neighbours, alignments surviving decontamination — is the only block of features that predicts contact accuracy, and it predicts every predictor.

Spearman ρ with MSA depth: #199 cooldown **0.67**, **#232 m2-p06 0.50**, ESMFold 0.46, ESMFold2 0.41, seq-KNN 0.37, Protenix + MSA 0.30, Protenix single-seq **-0.06**. All q < 0.001 and stable across eval-val and eval-test.

Everything else is noise by comparison. Length, relative contact order and fraction of long-range contacts are all |rho| <= 0.04 for MarinFold. The difficulty axes everyone assumes matter do not move our model.

We are the homology-dependent one

The hypothesis going in was that a single-sequence model should hold up where MSAs are thin. **The opposite is true.** Between the deepest quartile (7,413 to 19,393 sequences) and the shallowest (2 to 784, median 160), MarinFold loses **0.25** R-precision (0.631 -> 0.378). ESMFold2 loses 0.18 and Protenix-v2 + MSA loses 0.08. The gap to ESMFold2 widens as the family shrinks (rho = +0.18, p = 0.0014).

The same ordering appears in how *predictable* each predictor is from protein properties alone: cross-validated R-squared runs seq-KNN 0.86, #199 cooldown 0.49, #232 m2-p06 0.34, ESMFold 0.29, ESMFold2 0.24, Protenix + MSA about zero. The more a method leans on homology, the more its per-protein accuracy is a property of the protein rather than of the method. MarinFold sits closer to the KNN null than ESMFold2 does.

A sequence LM trained on a sequence database inherits that database's family statistics. That is the finding.

At the extreme -- the six proteins with ten or fewer MSA sequences -- the ordering changes but not in our favour. Protenix-v2 + MSA drops from 0.791 on Q1 to 0.410, and the KNN null to 0.139, so the MSA methods do lose their advantage. But MarinFold falls to 0.317, below its own Q1 average, while **ESMFold2 holds at 0.694**. The regime where "no MSA available" is the whole argument is exactly where a PLM-based predictor is strongest and we are weakest. Six proteins is a pointer, not a measurement.

Biology barely matters

Controlling for length, contact order and training identity, nothing in the biology block reaches half the strength of the homology block: UniProt domain count +0.21, bacterial +0.19, cytoplasmic -0.12, nuclear -0.11, viral -0.12, and membrane / secreted / enzyme all under 0.06. Secondary structure is small and points the unintuitive way -- sheet fraction +0.11, helix -0.09, so beta-rich proteins are marginally easier for us.

Viral survives the control, which matches #241 and #245: viral proteins are hard partly because their families are thin, and partly for a residue that is not just family size.

What to do about it

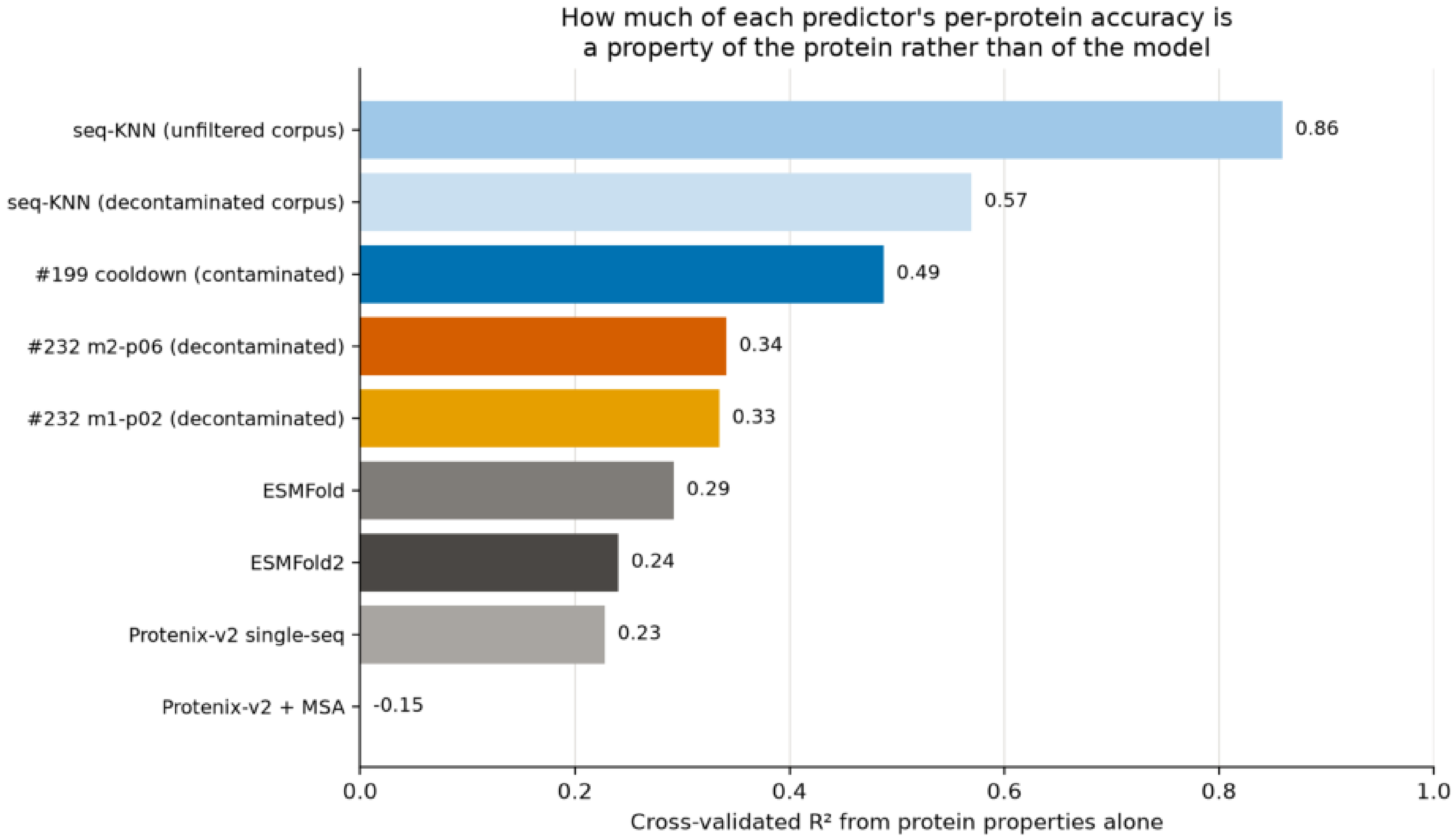
Training data. The lever is not more tokens of the same distribution, it is coverage of small families. Both corpus arms are built from clustered databases -- AFDB at struct-cluster level, ESM-Atlas at 40 % linclust -- which systematically downweight exactly the proteins we fail on. #241 found both arms miss viruses; this generalises it to thin families of any kind.

Evaluation. MSA depth belongs in the reporting as a stratum. A comparison on a set with median depth 3,000 says little about depth 100, and the ordering is not preserved: on the shallowest quartile MarinFold reaches 56 % of ESMFold2's score, on the deepest 74 %.

Open. This is correlational. A training-data intervention -- upsample small families, or train on a de-duplicated corpus -- is what would test causality.

explainable_variance.png

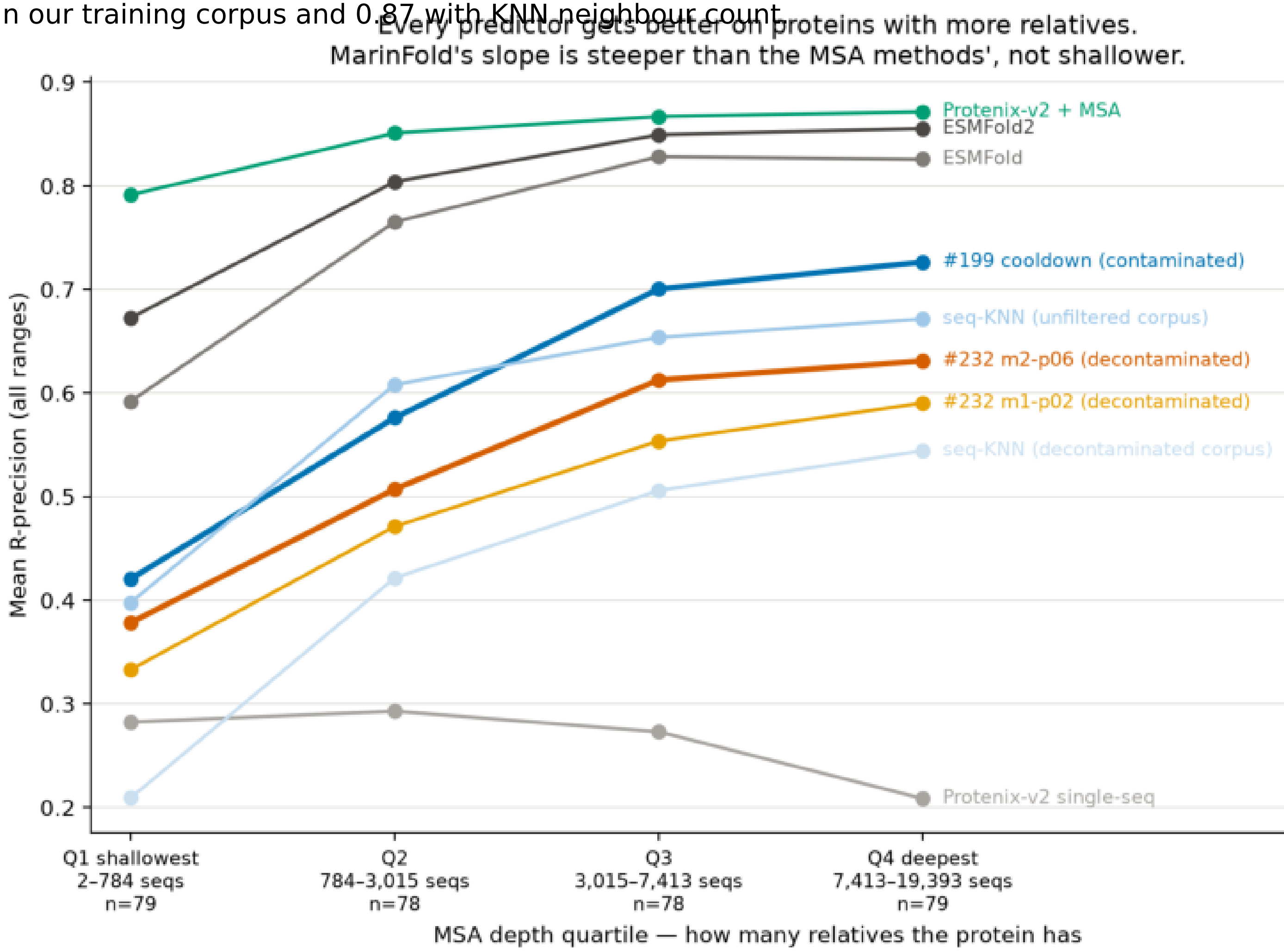
Cross-validated R^2 of the best of a ridge and a gradient-boosted model predicting each predictor's per-protein R-precision from 60 protein properties, 5-fold CV over 314 proteins. Negative values are clipped to zero.



$ plot_properties.py

family_abundance.png

Mean all-range R-precision by quartile of MSA depth over the 314 natural FoldBench monomers. Quartile boundaries are 784, 3,015 and 7,413 sequences; the set spans 2 to 19,393. MSA depth stands in for how many relatives a protein has — it correlates 0.80 with the number of homologs in our training corpus and 0.87 with KNN neighbour count.



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feature_heatmap.png

Spearman ρ between each protein property and each predictor's per-protein R-precision, pooled over the 314 natural monomers.

$ plot_properties.py